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Development of a Music-Based Productivity Tracker for Analyzing the Impact of Music on Focus During Study & Work Sessions

By:

Iyed Bayini

Members of the jury

Dr. Yassine Larbaoui	Lecturer
Dr. Anas Yaakoubi	Lecturer

Supervisor

Mr. Samir Jaafari

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I am grateful for the opportunity to study here, alongside the professors who have honored me with their knowledge, expertise, and appreciation for my hard work.

I also express my thanks to the members of the jury, who have evaluated my project.

And most importantly, to my family, relatives, and dearest of friends and loved ones, I have only boundless love for them, and gratitude for their presence in my life, especially when things seemed so unmanageable at times.

Declaration of Authorship

I hereby declare that the work presented in this project is entirely my own, except where otherwise referenced.

Abstract:

This project presents the development of a productivity tracker (dubbed Coda) that analyzes how different types of music affect user focus during productivity sessions. It enables users to complete timed sessions, select a music type, and provide a focus rating, with the collected data used to generate insights into productivity patterns for the users' benefit.

Coda was developed and will be deployed as a web application, and includes features like session tracking, statistical analysis, and data visualization. The application evaluates user performance across different music categories to identify patterns and trends that can then be turned into valuable insights for their use and benefit!

The research we have conducted on the matter of music's influence on individual focus indicates that music can have varying effects on concentration depending on its characteristics, with instrumental music usually inspiring better focus compared to lyrical or any form of high intensity music. This project demonstrates how data-driven analysis can be integrated into a productivity tool to equip users with personalized intelligence, helping them

achieve their goals by guiding them towards the appropriate kind of tunes to use!

Chapter 1: Overview, Description & Research

Introduction:

Our project revolves around the creation of a music-based productivity tracker, designed to analyze how different types of music can influence a user's concentration during study sessions. It allows users to perform timed sessions, select a music type, and rate their level of concentration upon completion via self-evaluation (putting trust on the user's opinion). The user's data is collected, then processed to generate meaningful insights regarding focus performance, which will allow them to notice patterns that they may have not seen on their own.

It was motivated by my desire to combine my three favourite fields: Music, Software Development, and data analytics, to create a solution that benefits people who find it challenging to function without music (such as myself), and to give them the opportunity to integrate it into their study & work times in a beneficial way.

Coda was developed using modern technologies and follows basic software development approach, including analysis, design, implementation, and testing. The key features include session creation and tracking, self-evaluated focus rating, statistics, and visual representations of performance across different music categories with charts and visible numbers. All of this enable users to observe patterns in their productivity and identify which types of music are most effective for maintaining their concentration, providing them with the necessary knowledge to optimize their work.

The results demonstrate that integrating data analysis into a simple productivity tool can provide valuable personalized insights. The system successfully transforms subjective user experiences into measurable data, supporting more informed decisions about study habits. This project highlights the potential of combining software development with data-driven analysis to address real-world productivity challenges that many people face today.

Background:

In recent years, maintaining focus has become increasingly challenging, particularly due to the growing number of distractions present in modern environments. As a result, many individuals seek methods to improve concentration and productivity, with music being one of the most commonly used tools, especially for the youth. Different genres such as lo-fi, classical, ambient sound, and white noise are frequently used with the intention of enhancing focus and reducing distractions.

Despite the widespread use of music for productivity, there is a lack of personalized and data-driven approaches that allow individuals to understand how different types of music affect their own concentration. Most existing solutions, such as general productivity applications or music streaming platforms, do not provide any meaningful insights into the relationship between music and focus, and don't offer much in that regard altogether. Instead, users rely on assumptions or general recommendations that may not be effective for them.

This creates a gap between the use of music as a productivity tool and the ability to measure its actual impact. Without structured tracking and analysis, users are unable to determine which types of music genuinely improve their focus, and are left fending for themselves, relying only on feeling and "vibes", which aren't always reliable. This highlights the need for a system that can collect, analyze, and present personalized data, enabling users to make informed decisions about their study habits and optimize their productivity to the best of their abilities.

The Problem:

Despite the widespread use of music as a tool to improve concentration during study and work sessions, there is no effective way for individuals to measure how different types of music impact their personal focus. Existing productivity tools and music platforms do not provide data-driven insights into the relationship between music and performance, leaving users to rely on assumptions rather than evidence.

This lack of personalized analysis creates uncertainty, as individuals cannot determine which types of music genuinely enhance or hinder their concentration if their feelings are the only metric they have. As a result, users may unknowingly adopt study habits that do not serve them, reducing their overall productivity.

Therefore, there is a need for a system that can track focus during study sessions, associate performance with specific music types, and provide meaningful analytical feedback. Such a system would enable users to make informed decisions and optimize their productivity based on measurable data.

Aim & Objectives:

The aim of this project is to design and develop a music-based productivity tracking system that analyzes how different types of music affect a user's focus during sessions and provides data-driven insights to improve productivity.

Its objectives are as follows:

- To design and implement a system that allows users to perform timed sessions.
- Enabling users to select and associate different types of music with each session
- To collect and store user data, including session length and focus ratings.
- Calculating and displaying metrics such as total focus time, average rating, and session counts.
- Analyzing performance across different music types to identify patterns.
- To provide users with insights such as the most effective, disruptive, and frequently used music
- To visualize data through charts and summaries for a better understanding of productivity trends through consistent usage of the software.
- Evaluating the effectiveness of the system in helping users optimize their study habits.

Scope and Limitations:

The scope of this project focuses on designing and developing a music-based productivity tracking system that allows users to complete timed study/work sessions

and evaluate their level of focus. It enables users to select a type of music for each session and provides a focus rating upon completion.

The project is developed as a web-based application using modern technologies and focuses primarily on functionality, data tracking, analysis, and feedback.

The application includes features such as session tracking, calculation of total focus time, average focus ratings, and the analysis of performance. Additionally, the system provides vital insights, such as the most and least effective, and most frequently used music categories.
Visual representations in the form of charts are included to help users better understand their patterns.

Despite its functionality, the system does have its limits.

Firstly, the application relies on user self-assessment to determine focus levels, introducing subjectivity and thus potentially affecting the accuracy of the results.

Secondly, the system does not integrate with external music streaming platforms for the time being, meaning that users must manage their music separately.

Analysis is limited to the data provided by the user, without incorporating external factors such as environment, task complexity, or state of mind, which may also influence focus. The system is designed for individual use and does not include collaborative features of any sort.

Finally, due to resource constraints, the app focuses on core functionality rather than advanced features, such as real-time tracking, recommendations, or large-scale deep analysis.

Project Overview:

Coda, the Music-Based Productivity Tracker, is a web-based application designed to help users improve their focus by analyzing the impact of different types of music in their hours of focus. The system allows users to perform timed sessions, select a music type, and provide a focus rating upon completion.

Each session is recorded and stored, including information such as duration, selected music category, and user-assigned focus level. This data is then processed

to generate information that will then serve as valuable intelligence (focus time, rating, session numbers)

The system also analyzes performance across different music types, enabling users to identify which types of music are most effective, least effective, and most frequently used. These are all presented through statistical summaries and visual representations to make the data easier to understand.

The application is designed with a simple and intuitive user interface, allowing users to easily start sessions, input feedback, and view results. By combining session tracking with data analysis, the system provides a personalized approach to productivity, helping users make informed decisions about their study habits.

Technical Review:

The development of this project required the use of several technologies to ensure a functional, interactive, and user-friendly system. The selection of these technologies was based on simplicity, performance, and suitability for building a data-driven web application.

Frontend Framework:

The application was developed using Angular, a widely used frontend framework that enables the creation of dynamic and responsive web applications. Angular was chosen due to its component-based architecture, which allows the system to be organized into reusable and manageable parts, perfect for maintainability and scalability.

Programming Language:

TypeScript was used as the primary programming language. TypeScript extends JavaScript by adding static typing, which improves code reliability and reduces potential runtime errors. Its integration with Angular makes it a suitable choice for building structured and maintainable applications, such as this one.

Data Storage:

The system uses a client-side data storage approach to manage session data. This allows for quick data retrieval and simplifies the development process without requiring a backend server. While this approach is sufficient for a single-user application, it also highlights potential areas for future improvement, such as database integration and secure storage.

Data Visualization:

Chart.js was used to implement data visualization features within the system. This library enables the creation of interactive and visually appealing charts, which are used to display average focus levels across different music types. The use of visual representations enhances user understanding of productivity patterns, and simplifies comprehension even further.

Development Tools:

The project was developed using Visual Studio Code as the primary development environment. Version control was managed using Git, allowing for tracking changes and maintaining code integrity throughout the development process.

Overall, the selected technologies provided a balance between simplicity and functionality, enabling the successful implementation of a system that combines session tracking with data analysis in a user-friendly manner.

Design Decisions and Justification

The design and development of the system involved several key decisions that were made to balance functionality, simplicity, and scalability. Each decision was influenced by the project's objectives, technical constraints, and the need to create a user-friendly and effective application.

One of the primary design choices was the use of a client-side architecture. By storing and processing data directly within the browser using local storage, the system avoids the need for a backend server or database. This significantly reduces complexity and development time, making the system easier to implement and maintain. While this approach limits the application to single-user usage, it is appropriate for the scope of this project, where the primary goal is to demonstrate functionality and data analysis rather than large-scale deployment.

Another important decision was the selection of Angular as the frontend framework. Angular's component-based architecture allows the application to be divided into distinct, reusable modules, such as the session component and the dashboard component. This separation of concerns improves code organization and makes the system easier to extend in the future. For example, additional features such as user

authentication or database integration could be implemented without significantly restructuring the existing codebase.

The use of TypeScript further supports maintainability and reliability. By introducing static typing, TypeScript reduces the likelihood of runtime errors and improves code readability. This is particularly beneficial in a data-driven application, where accurate handling of variables such as session duration and focus ratings is essential.

The decision to use Chart.js for data visualization was based on its simplicity and effectiveness. The library provides a straightforward way to create interactive charts that clearly represent performance data. Visualizing data in this way allows users to quickly identify patterns and trends, which may not be immediately apparent from numerical values alone. This aligns with the project's goal of making data accessible and easy to interpret.

From a user experience perspective, simplicity was prioritized throughout the design process. The interface was intentionally kept minimal, with a clear workflow that guides the user from starting a session to viewing results. This reduces cognitive load and ensures that users can focus on their tasks rather than navigating the application.

Additionally, the system was designed with extensibility in mind. Although the current implementation is relatively simple, the modular structure allows for future enhancements such as database integration, multi-user support, and more advanced analytical features. This forward-thinking approach ensures that the system can evolve beyond its initial scope.

The design decisions made in this project reflect a balance between practicality and potential. By prioritizing simplicity, functionality, and scalability, the system successfully achieves its objectives while maintaining the flexibility for future development.

Literature Review

Lyrical Music: A Source of Distraction

A recurring trend in the literature is that lyrical music tends to disrupt cognitive performance, particularly in tasks that require comprehension and reading. Salamé and Baddeley (1989) demonstrated that lyrical music interferes with short-term memory, as the presence of lyrics introduces additional verbal information that competes for cognitive resources. This phenomenon can be explained through the

concept of cognitive load, where the brain has a limited capacity for processing information. When both the task and the music require linguistic processing, interference occurs, leading to reduced performance.

Similarly, Davisson (2021) found that participants performing reading-based tasks performed worse when exposed to lyrical music than when exposed to instrumental music or silence. This suggests that the processing of lyrics directly conflicts with the cognitive demands of reading and comprehension, reinforcing the idea that not all forms of background music are beneficial for productivity.

Perham and Currie (2014) further observed that individual music preferences do not necessarily mitigate this effect. Their findings suggest that even when participants preferred a particular genre, exposure to high-intensity vocal music still negatively impacted their concentration and task performance. This indicates that the disruptive effect of lyrics may outweigh personal preference in cognitively demanding tasks. Therefore, while individuals may perceive certain music as helpful, objective performance measures may reveal the opposite.

Instrumental Music: A Potential Cognitive Aid

In contrast, several studies suggest that instrumental music can enhance focus and improve cognitive performance under certain conditions. Lin et al. (2021) found that students exposed to instrumental and lo-fi music demonstrated improved concentration and productivity compared to those working in silence. This improvement may be attributed to the absence of linguistic interference, allowing the brain to remain engaged without overloading its processing capacity.

Additionally, participants in Lin et al.'s study exhibited higher heart rate variability (HRV), indicating a state of relaxed alertness that is conducive to sustained attention. This physiological response suggests that instrumental music may help regulate emotional and cognitive states, creating an environment that supports focused work.

Angwin et al. (2017) similarly argued that instrumental music can reduce anxiety and promote a calm mental state, thereby enhancing task performance. This reduction in stress may be particularly beneficial in academic or high-pressure environments, where anxiety can negatively affect concentration. Together, these findings suggest that music without lyrics may provide an optimal balance between stimulation and distraction, supporting both engagement and efficiency.

Loudness, Genre, and Individual Differences

The impact of music on focus is not solely determined by the presence of lyrics, but also by factors such as loudness, genre, and individual differences. Wang (2024) observed that loud and energetic music, such as pop, negatively affected concentration, particularly during reading and comprehension tasks. This suggests that overstimulation can impair cognitive performance, even when the music does not contain lyrics.

Furthermore, Kampfe et al. (2011) found that individuals' perceptions of music as helpful do not always align with actual performance outcomes. Many participants reported that music improved their focus, despite objective measurements indicating decreased performance. This highlights a disconnect between subjective experience and measurable productivity, emphasizing the importance of data-driven analysis when evaluating study habits.

However, Ferreri et al. (2013) proposed that music can enhance memory and performance when it aligns with the context or "atmosphere" of a task. For example, calm and repetitive music may support tasks requiring sustained attention, while more stimulating music may be beneficial for creative activities. This suggests that the relationship between music and focus is complex and influenced by multiple interacting factors, including task type, personal preference, and environmental context.

Research Gap

Although existing studies provide valuable insights into how music influences cognitive performance, there remains a lack of practical systems that allow individuals to measure these effects in real-world scenarios. Most research is conducted in controlled laboratory environments, which may not accurately reflect the variability and unpredictability of everyday study conditions.

Additionally, existing productivity tools often focus on time management rather than cognitive performance, and they rarely integrate personalized data analysis with music usage. As a result, users are left to rely on assumptions and subjective judgment when selecting music for studying.

This highlights the need for a system that enables users to track their focus across different types of music and analyze their performance over time. By collecting user-specific data and transforming it into meaningful insights, such a system can bridge the gap between theoretical research and practical application. The development of a tool that combines data tracking, analysis, and visualization can provide users with a clearer understanding of how music affects their productivity, ultimately supporting more informed and effective study habits.

Research Methodology:

Coda was tested by multiple users who conducted sessions with varying kinds of music. Said participants included students who work whilst listening to music and claim to struggle to find the appropriate genres to achieve peak concentration.

Participants were asked to do the following:

1. Open the app and select a music type
2. Complete a full session to their liking
3. Rate their focus from 1-5 at the end
4. And to repeat a few more times

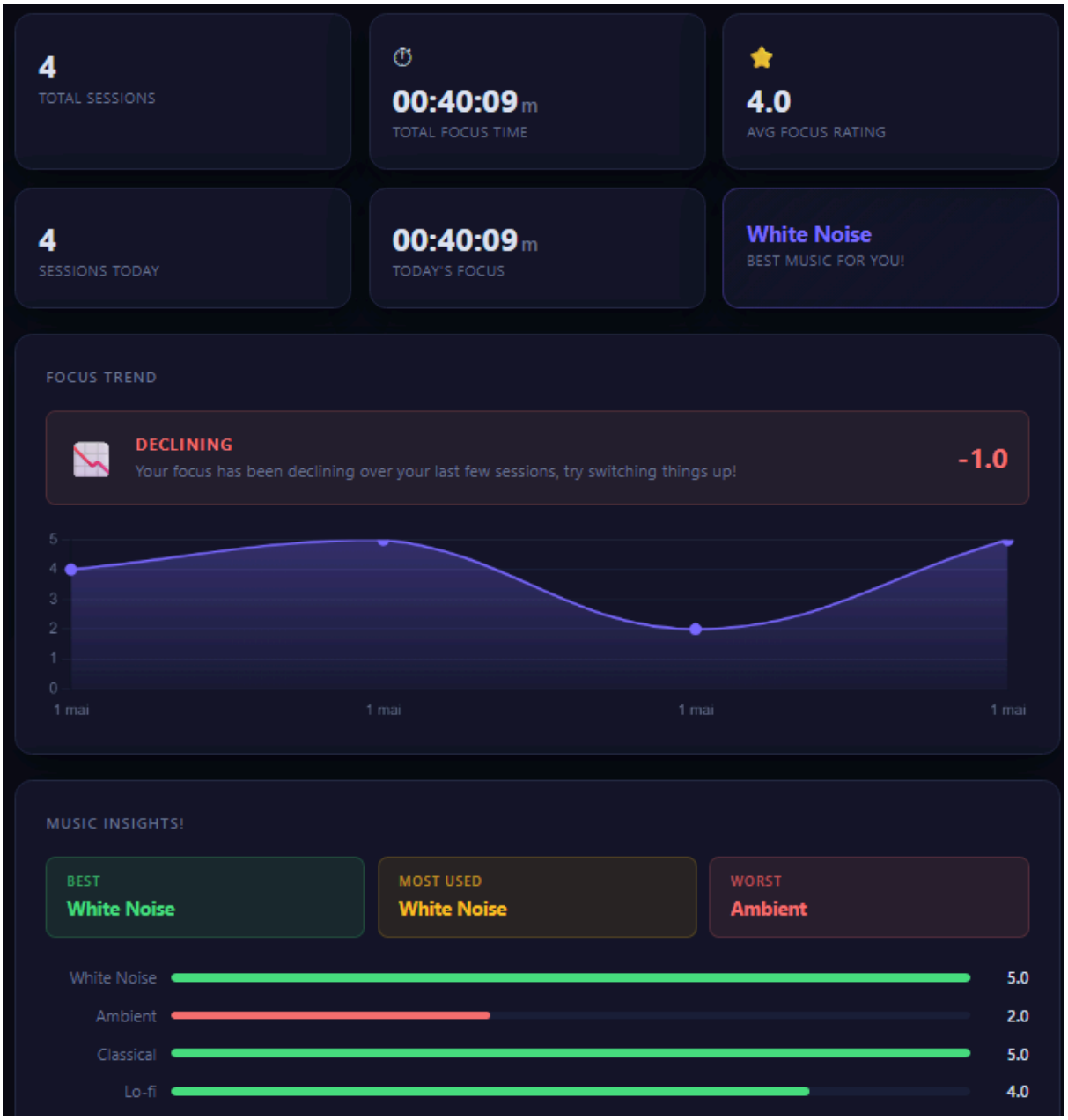
The data collected during testing included session duration, selected music type, and user-assigned focus ratings. This data was then processed by the system to calculate average focus levels, identify performance patterns, and determine the most effective music types for each user.

This methodology enabled the collection of consistent, comparable data, allowing meaningful analysis of how different types of music influence focus and productivity.

Results & Findings:

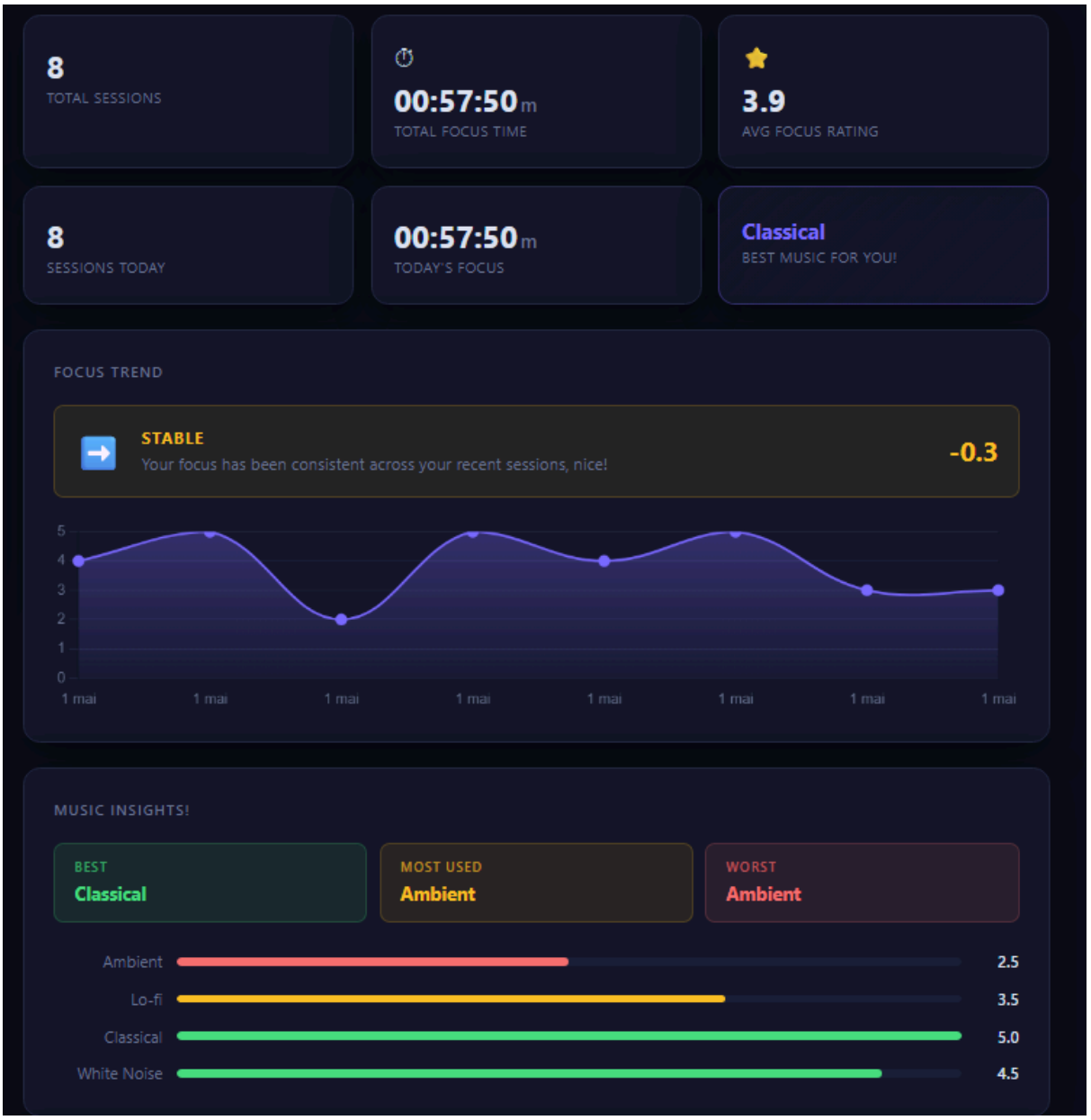
The results obtained from testing are presented below. Data collected from eight sessions was analyzed to identify patterns in focus across different kinds of music.

Phase 1: First four sessions:



Here, classical music and white noise achieved the highest average of 5.0, with Lo-fi at 4.0 and Ambient sound at a mere 2.0, not providing the effect that is desired.

Phase 2: Last four sessions



In this figure however, we can see that focus has remained rather consistent across sessions, and that classical music has cemented itself as the best genre for this particular user, followed by white noise.

Addendum:

The testing phase produced a set of results that demonstrate how different types of music influence focus during study sessions. A total of eight sessions were conducted across four different music types, allowing patterns in user performance to be observed.

The results indicate that Classical music achieved the highest average focus rating of 5.0, making it the most effective music type for maintaining concentration. White

Noise also performed strongly, with an average focus rating of 4.5, suggesting that non-lyrical and low-distraction audio supports focus effectively.

Lo-fi music produced a moderate average focus rating of 3.5, indicating that while it can support concentration, it may not be as effective as more neutral audio types. In contrast, ambient music resulted in the lowest average focus rating of 2.5, suggesting that it may not be suitable for tasks requiring sustained attention in this context.

These findings are consistent with our literature review, highlighting that instrumental and non-lyrical music tends to enhance cognitive performance, while other forms of music may be less effective depending on their characteristics.

The app also identified Classical music as the most effective category, while Ambient music was both the least effective and most frequently used. This suggests that user preference does not always align with actual performance outcomes.

Trend analysis indicated that user focus remained relatively stable across sessions, with only minor variation. This suggests a consistent level of performance over time, without significant improvement or decline.

The use of visual representations, including charts and trend indicators, made it easier to interpret these results and identify patterns across sessions.

the findings, however, are subject to certain limitations. The focus ratings were self-reported, introducing subjectivity into the mix. Also, the sample size was relatively small, which limits the generalizability of the results. External factors such as environment, mood, and task type were also not controlled.

Overall, the results demonstrate that the system can capture meaningful data and generate useful insights. Findings support the idea that different types of music can have a measurable impact on focus, and that a data-driven approach such as this can serve users in optimizing their productivity and achieve better results as a consequence.

Chapter 2: System Analysis and Design

System Analysis:

Coda is designed to provide users with a structured and systematic way to monitor and evaluate their focus during study sessions while listening to different types of music. The primary objective of the system is to enable the collection and analysis of session data in order to identify patterns in user performance and productivity.

The application records key session attributes such as duration, selected music type, and user-assigned focus rating. These variables form the foundation of the system's analytical capabilities. By capturing both quantitative data (session duration) and qualitative data (focus rating), the system is able to provide a more comprehensive view of user performance.

The collected data is processed to compute metrics such as average focus per music category, total study time, and overall performance trends across multiple sessions. These metrics allow users to move beyond subjective judgment and instead rely on measurable data when evaluating their study habits.

Based on this analysis, the system generates insights that help users understand which types of music are most effective for maintaining their concentration. These insights are presented through both numerical summaries and visual representations, such as charts, allowing for clear and intuitive interpretation. This combination of data and visualization enhances usability and ensures that users can easily understand the results.

The system operates entirely on the client side, using local storage to manage session data. This design decision simplifies the system architecture by removing the need for a backend server while still ensuring that user data persists across sessions. It also enhances accessibility, as users can interact with the system directly through a web browser without requiring additional setup.

Overall, the system is designed to transform raw user input into meaningful feedback, supporting more informed and personalized study habits. By bridging the gap between data collection and interpretation, the system provides practical value to users seeking to improve their productivity.

Functional Requirements:

The functional requirements define the core operations that the system must perform in order to meet its objectives. These requirements ensure that the system is capable of capturing, processing, and presenting user data effectively.

The app shall allow users to start and manage timed focus sessions, providing a structured environment for studying. Users must be able to select a type of music before starting a session, allowing the system to associate performance data with specific audio conditions.

It shall automatically track the duration of each session, ensuring accurate measurement of study time without requiring manual input. Users shall also be able to stop or reset a session at any time, providing flexibility and control over their workflow.

Upon completion of a session, the system shall allow users to provide a focus rating. This rating is a key input used in subsequent analysis, as it reflects the user's perceived level of concentration.

It will store session data, including start time, end time, duration, selected music type, and focus rating. This data must be retrievable for display and analysis purposes. The system shall calculate total focus time, total number of sessions, and the average focus rating across sessions.

In addition, Coda will analyze focus performance based on different music types, identifying patterns and trends. It shall determine and display the best-performing, worst-performing, and most frequently used music categories.

The system shall present analytical data through visual representations such as charts, making it easier for users to interpret results. It shall also provide insights into user performance trends over time, allowing users to monitor changes in their productivity.

Non-Functional Requirements

The non-functional requirements define the quality attributes of the system, ensuring that it operates efficiently, reliably, and in a user-friendly manner.

The system shall provide a simple and intuitive user interface, allowing users to easily navigate and interact with the application without prior training. Usability is a key consideration, as the effectiveness of the system depends on how easily users can engage with it.

The system shall respond quickly to user actions, ensuring that session tracking, data updates, and navigation occur without noticeable delay. Performance efficiency is important to maintain a smooth user experience.

The system shall ensure data persistence by storing session data locally, allowing users to retain their progress even after refreshing or reopening the application. This eliminates the need for repeated data entry and enhances user convenience.

Accuracy is another critical requirement. The system shall maintain precision in all calculations, including session duration, average focus ratings, and music-based analysis. Reliable data processing is essential for generating meaningful insights.

The system shall also be reliable, ensuring that user data is consistently stored and retrieved without loss or corruption. This builds user trust and ensures the integrity of the analysis.

Additionally, the system shall be accessible through a web browser without requiring installation, making it easy for users to access and use the application across different devices.

Finally, the system shall be scalable in design, allowing for future enhancements such as database integration, more advanced analytics, and multi-user functionality. This ensures that the system can evolve beyond its initial implementation.

System Design

The system is designed using a modular, component-based architecture to ensure clarity, maintainability, and scalability. The application is built using Angular, which supports the separation of concerns through the use of components and services. This architectural approach allows different parts of the system to be developed, tested, and maintained independently.

The system consists of two main components. The focus session component is responsible for managing the timer, handling user interactions, and collecting session data such as duration, selected music type, and focus rating. This component serves as the primary interface for user input.

The dashboard component is responsible for displaying processed data, including statistics, insights, and visual representations of user performance. It transforms raw session data into meaningful outputs that users can interpret and act upon.

A dedicated service layer is used to manage session data. The session service handles storing, retrieving, and updating session information, ensuring consistency across the application. By centralizing data management, the system reduces redundancy and improves maintainability.

The service layer also manages data persistence by saving session data in the browser’s local storage. This allows the application to retain user data across sessions without requiring external storage solutions.

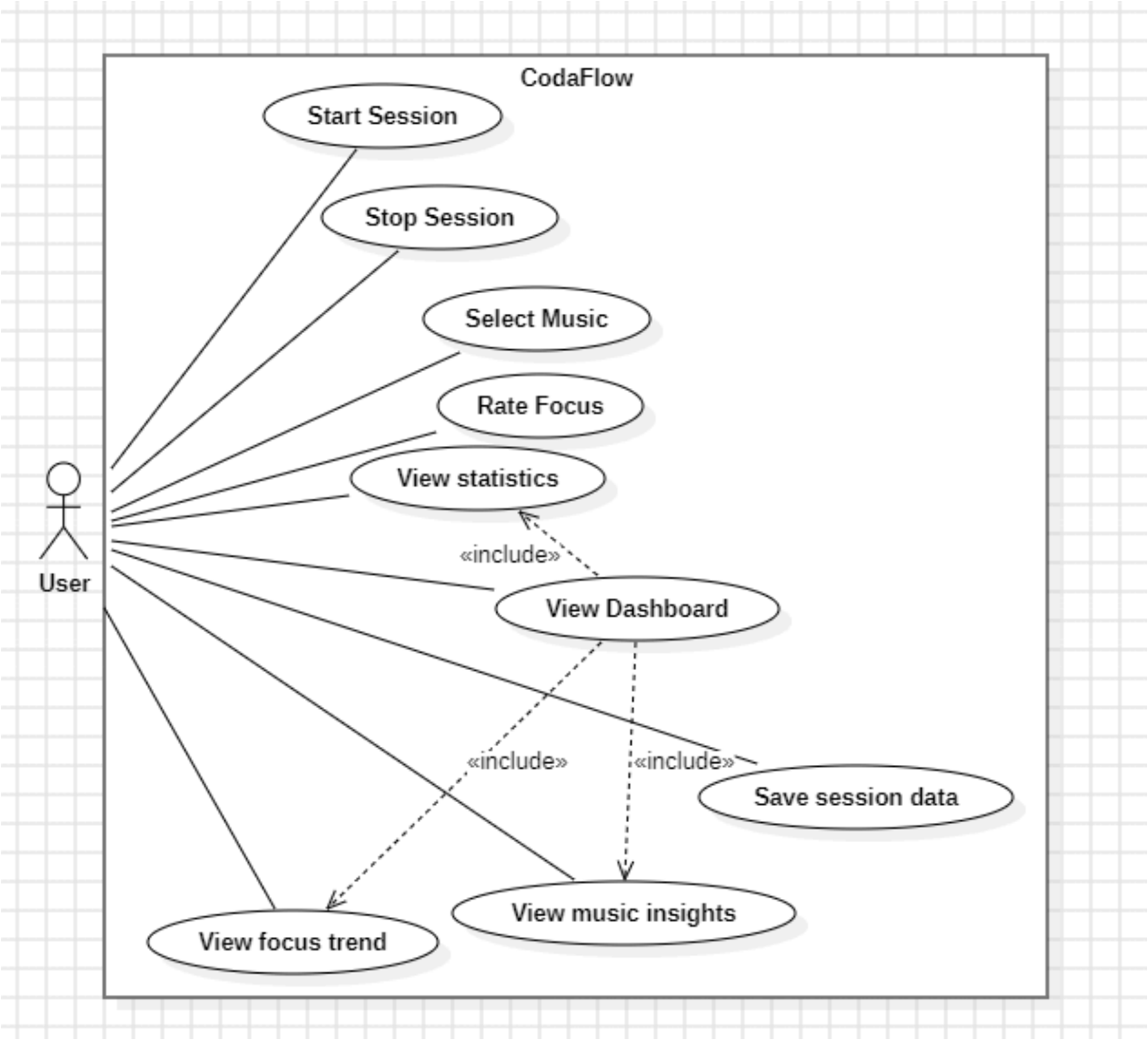
The system follows a clear and logical data flow. User actions generate session data, which is passed to the service layer for storage. This data is then retrieved and processed to calculate metrics such as average focus, most effective music types, and performance trends. The processed results are displayed through the dashboard interface in both numerical and visual formats.

The design prioritizes simplicity and usability, ensuring that users can easily interact with the system and understand their performance. At the same time, the modular structure supports future enhancements, such as integrating a database or adding multi-user functionality, making the system adaptable and scalable.

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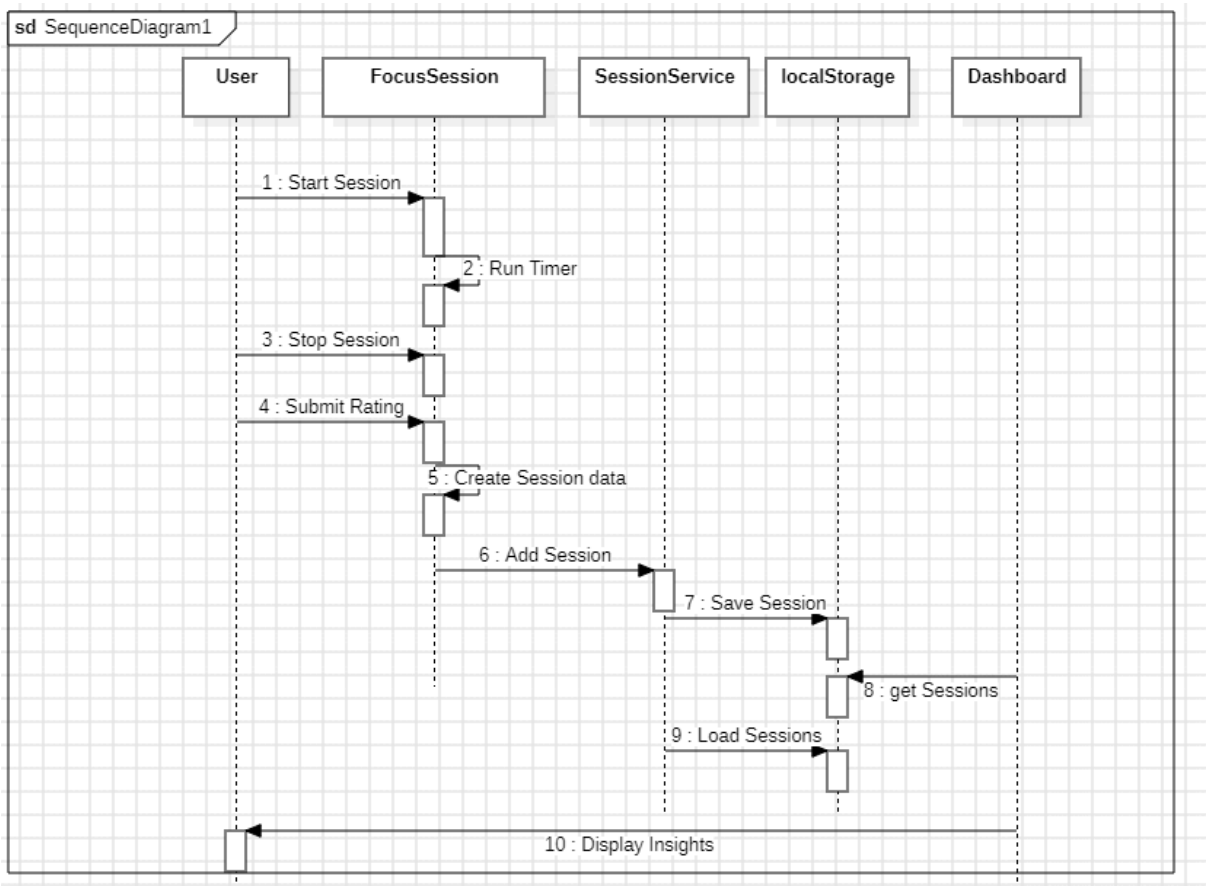
Diagrams:

Use Case:



This diagram demonstrates what the actors (in this case, users of the app) are capable of doing within the bounds of the software. As we can see, users have the ability to select their desired music type, start and stop their session at will, then rate their focus level at the end of them. They may then check the dashboard out to take a look at their statistics, insights, and trends that have been provided to them.

Sequence Diagram:



This diagram showcases the flow of operations within Coda. What happens from the beginning all the way to the end when a user goes through the application's functions. The user starts a session, and by consequence, a timer along with it. They may then stop it whenever they're finished, and submit their self-evaluated focus rating. The software will then create the appropriate session to be stored in the user's browser local storage. When that is done, the relevant insights will be displayed on the dashboard for the user's viewing.

Chapter 3: Implementation, Testing and Demonstration

Implementation:

Focusing on key functionalities and the logic used to develop the application. Code snippets will be presented to demonstrate how session management, data storage and analysis, and visualization are achieved.

Session Creation:

```
submitSurvey() {
  if (this.focusRating === null) return;

  this.surveySubmitted = true;

  const sessionResult: Session = {
    startTime: this.startTime!,
    endTime: this.endTime!,
    duration: this.totalSeconds - this.remainingSeconds,
    focusRating: this.focusRating,
    musicPlayed: this.selectedMusic,
    mode: 'pomodoro'
  };

  this.sessionService.addSession(sessionResult);
  console.log('Session result:', sessionResult);
}
```

This function is responsible for creating a session record after the user completes a session.
It collects relevant data such as start time, end time, duration, selected music type, and the user's focus rating.

Once the session object is created, it's passed to the session service for storage. This process ensures that input is structured and recorded, forming the foundation for further analysis.

Data Storage:

```typescript
import { Injectable } from '@angular/core';
import { Session } from '../models/session.model';

const STORAGE_KEY = 'codaflow_sessions';

@Injectable({
  providedIn: 'root'
})
export class SessionService {
  private sessions: Session[] = this.load();

  addSession(session: Session): void {
    this.sessions.push(session);
    this.save();
  }

  getSessions(): Session[] {
    return this.sessions;
  }

  private save(): void {
    localStorage.setItem(STORAGE_KEY, JSON.stringify(this.sessions));
  }

  private load(): Session[] {
    const raw = localStorage.getItem(STORAGE_KEY);
    if (!raw) return [];
    return (JSON.parse(raw) as any[]).map(s => ({
      ...s,
      startTime: new Date(s.startTime),
      endTime: new Date(s.endTime)
    }));
  }
}
```

The system uses browser local storage for data persistence, ensuring that user information remains available even after refreshing or reopening the application.

The save() function converts the session data into JSON format and stores it locally. The load() function retrieves this data and reconstructs it into usable objects. Since local storage stores data as strings, date values are converted back into Date objects during retrieval. This ensures that time-based operations such as formatting, sorting, and duration calculations continue to function correctly.

This approach provides a simple and efficient method of data persistence without requiring a database or backend system.

Data Analysis and Insight Generation:

```
getAverageFocusByMusicType(music: string): number {
  const filtered = this.sessions.filter(s => s.musicPlayed === music);
  if (filtered.length === 0) return 0;

  const total = filtered.reduce((sum, s) => sum + s.focusRating, 0);
  return total / filtered.length;
}

get bestMusicType(): string {
  if (this.uniqueMusicType.length === 0) return 'None';

  let best = this.uniqueMusicType[0];
  let bestRating = this.getAverageFocusByMusicType(best);

  for (const music of this.uniqueMusicType) {
    const score = this.getAverageFocusByMusicType(music);
    if (score > bestRating) {
      best = music;
      bestRating = score;
    }
  }

  return best;
}
```

This function is responsible for analyzing session data by calculating the average focus rating for a specific music type. It filters the stored sessions based on the selected music category and computes the mean value of their focus ratings.

This process transforms raw session data into meaningful numerical metrics, which serve as the foundation for further analysis and comparison between different music types.

```
get mostUsedMusicType(): string {
  if (this.sessions.length === 0) return 'None';

  const counts: { [key: string]: number } = {};

  for (const s of this.sessions) {
    const music = s.musicPlayed || 'None';
    counts[music] = (counts[music] || 0) + 1;
  }

  let mostUsed = this.uniqueMusicType[0];
  let maxCount = counts[mostUsed];

  for (const music in counts) {
    if (counts[music] > maxCount) {
      mostUsed = music;
      maxCount = counts[music];
    }
  }

  return mostUsed;
}

get worstMusicType(): string {
  if (this.uniqueMusicType.length === 0) return 'None';

  let worst = this.uniqueMusicType[0];
  let worstScore = this.getAverageFocusByMusicType(worst);

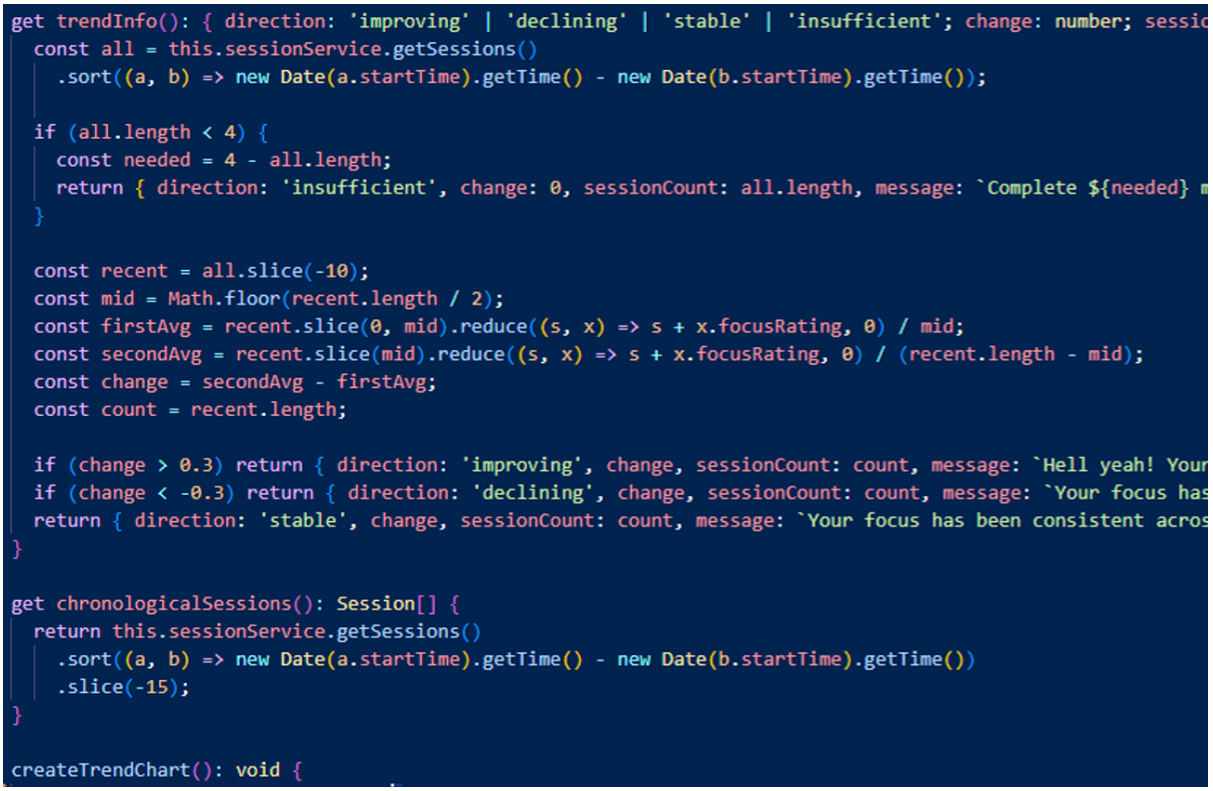
  for (const music of this.uniqueMusicType) {
    const score = this.getAverageFocusByMusicType(music);
    if (score < worstScore) {
      worst = music;
      worstScore = score;
    }
  }
```

The system generates insights based on the analyzed data to help users understand their performance. Functions such as determining the best, worst, and most frequently used music types are implemented by comparing calculated averages and counting session occurrences.

For example, the best music type is identified by selecting the category with the highest average focus rating, while the most used type is determined by counting how often each category appears in the session's data.

These insights provide users with clear and actionable feedback, allowing them to make informed decisions about which types of music improve their focus.

Trend Analysis:



```
get trendInfo(): { direction: 'improving' | 'declining' | 'stable' | 'insufficient'; change: number; sessio
  const all = this.sessionService.getSessions()
    .sort((a, b) => new Date(a.startTime).getTime() - new Date(b.startTime).getTime());

  if (all.length < 4) {
    const needed = 4 - all.length;
    return { direction: 'insufficient', change: 0, sessionCount: all.length, message: `Complete ${needed} m
  }

  const recent = all.slice(-10);
  const mid = Math.floor(recent.length / 2);
  const firstAvg = recent.slice(0, mid).reduce((s, x) => s + x.focusRating, 0) / mid;
  const secondAvg = recent.slice(mid).reduce((s, x) => s + x.focusRating, 0) / (recent.length - mid);
  const change = secondAvg - firstAvg;
  const count = recent.length;

  if (change > 0.3) return { direction: 'improving', change, sessionCount: count, message: `Hell yeah! Your
  if (change < -0.3) return { direction: 'declining', change, sessionCount: count, message: `Your focus has
  return { direction: 'stable', change, sessionCount: count, message: `Your focus has been consistent acros
}

get chronologicalSessions(): Session[] {
  return this.sessionService.getSessions()
    .sort((a, b) => new Date(a.startTime).getTime() - new Date(b.startTime).getTime())
    .slice(-15);
}

createTrendChart(): void {
```

This function analyzes changes in user performance over time by comparing recent session data. It first retrieves and sorts all sessions chronologically, then focuses on a subset of the most recent sessions, requiring a minimum of four sessions to provide trends.

The data is divided into two groups, representing earlier and more recent sessions. The average focus rating is calculated for each group, and the difference between these averages determines the overall trend.

Based on this comparison, the system classifies user performance as improving, declining, or stable. This allows users to understand how their focus evolves over time and provides meaningful feedback beyond simple statistics.

Data Visualization:

```
// THE chart
createChart(): void {

  const labels = this.uniqueMusicType;
  const data = labels.map(music => this.getAverageFocusByMusicType(music));

  if (labels.length === 0) return;

  const backgroundColors = data.map(value => {
    if (value >= 4) return 'rgba(75, 192, 192, 0.6)';
    if (value >= 3) return 'rgba(255, 206, 86, 0.6)';
    return 'rgba(255, 99, 132, 0.6)';
  });

  // destroys old chart
  if (this.chart) {
    this.chart.destroy();
  }

  // creates chart
  this.chart = new Chart(this.chartRef.nativeElement, {
    type: 'bar',
    data: {
      labels: labels,
      datasets: [{
        label: 'Average Focus',
        data: data,
        backgroundColor: backgroundColors,
        borderWidth: 1
      }]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      scales: {
        y: {
          beginAtZero: true,
          suggestedMax: 5
```

The system uses Chart.js to visually represent analyzed data, allowing users to easily interpret their performance. A bar chart is used to display the average focus rating for each music type, providing a clear comparison between different categories.

The chart is dynamically generated using processed data, where each bar corresponds to a specific music type and its associated average value. Visual elements such as colors and scaling are used to enhance readability and emphasize performance differences.

By transforming numerical data into graphical form, user understanding is improved, enabling quicker identification of patterns and trends in productivity.

Testing:

Coda was tested to ensure that all functionalities operated correctly and met the defined requirements. Testing focused on verifying the core features of the application, including session management, data storage, analysis, and visualization.

Functional testing was conducted by performing multiple focus sessions and confirming that session data, like duration, selected music type, and focus rating, was correctly recorded. It was also tested to ensure that stored data persisted after refreshing the page.

Data analysis features were tested by verifying that calculations such as average focus ratings, best and worst music types, and usage frequency were accurate and consistent. Trend analysis was also tested to ensure that changes in user performance over time were correctly identified.

The chart visualization was tested to confirm that data was displayed accurately and updated dynamically as new sessions were added.

Coda performed reliably overall, with all key functionalities working as expected and no major issues identified during testing.

Demonstration:

This section presents the final system interface and demonstrates the key features of the application.

Figure 1: Dashboard Interface
The dashboard provides an overview of user performance, displaying key statistics such as total sessions, total focus time, and average focus rating. It also highlights insights, including the most effective, most used, and least effective music types. This allows users to quickly understand their productivity patterns.

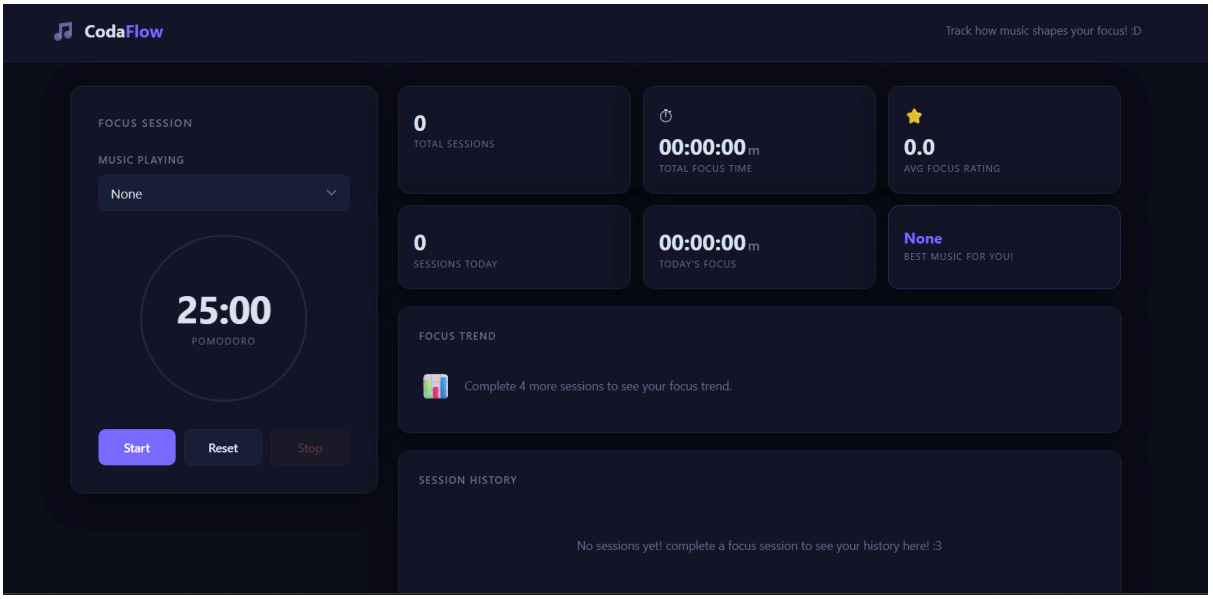
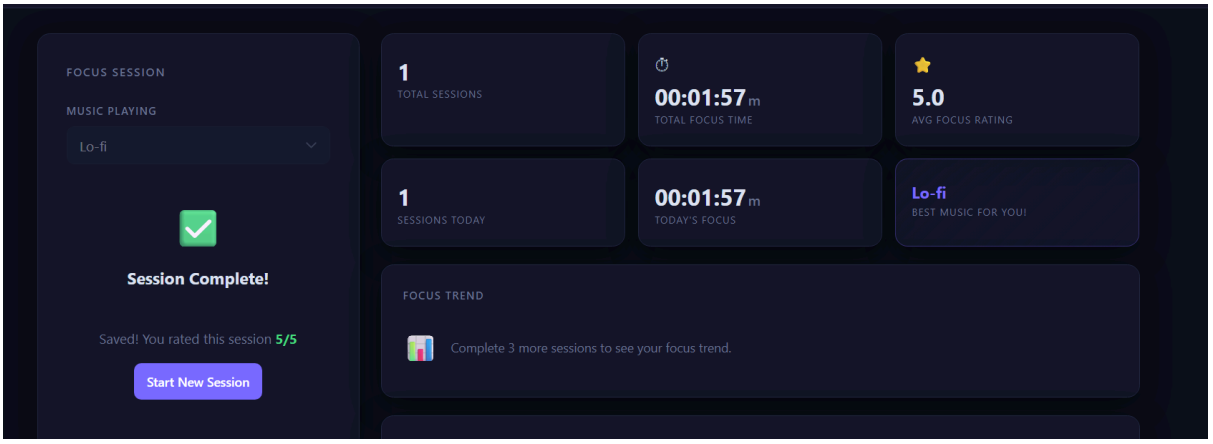


Figure 2: Focus Session Interface
The focus session interface enables users to start and manage timed sessions. Users can select a music type before beginning a session, and it tracks the session duration automatically. This interface serves as the primary interaction point for data collection.



Figure 3: Session Completion and Rating
At the end of each session, users are prompted to rate their level of focus on a scale from 1 to 5. This rating is stored along with session data and forms the basis for all subsequent analysis.



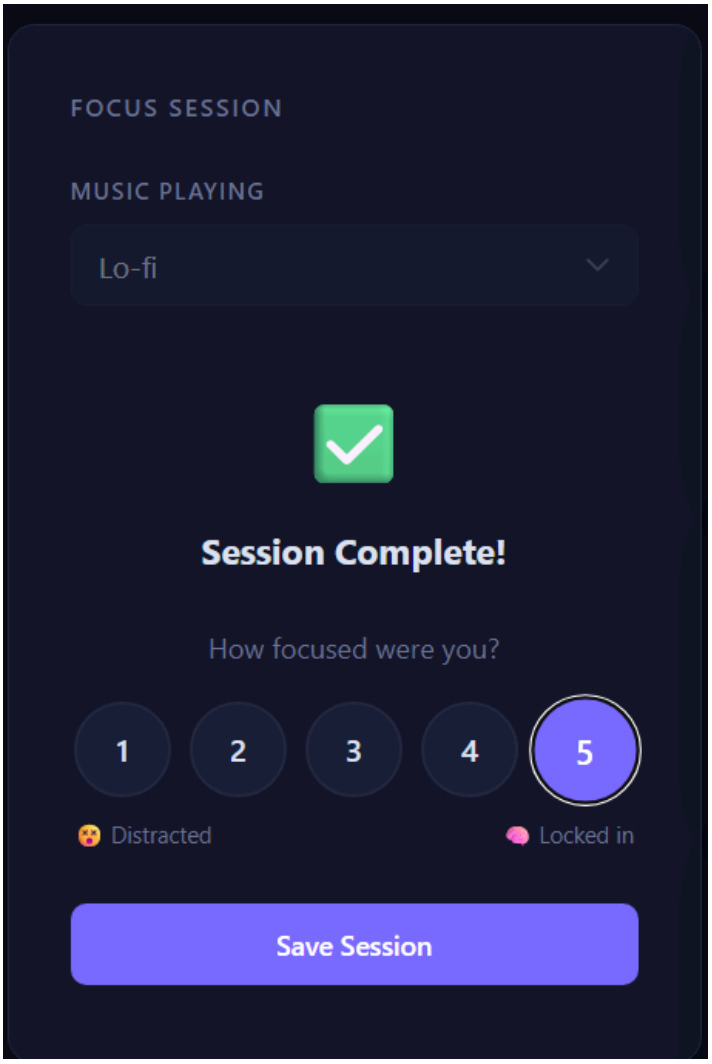


Figure 4: Insights Display
The system generates insights based on collected data, identifying the best, most used, and worst-performing music types. These insights help users understand which types of music are most effective for maintaining concentration.

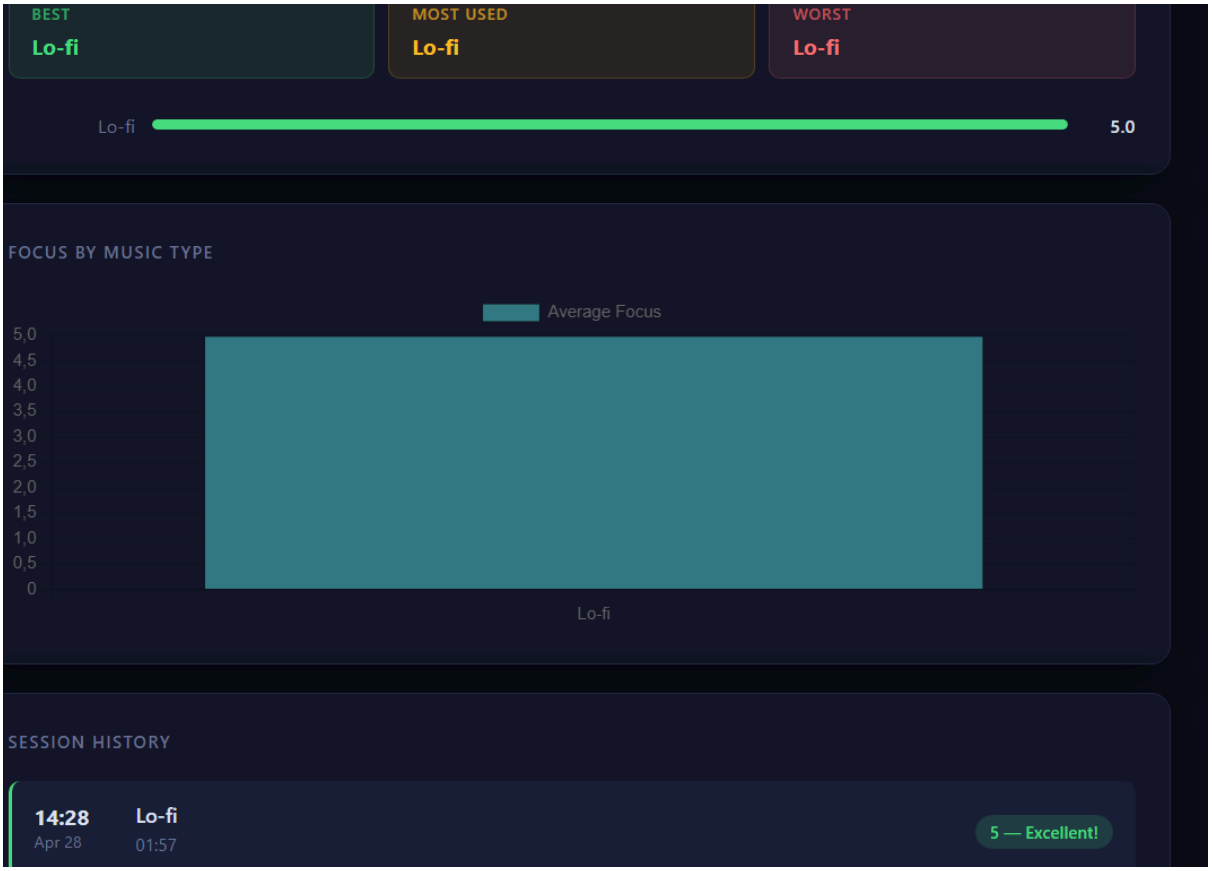


Figure 5: Focus Trend Analysis
The trend analysis feature visualizes changes in user performance over time. It identifies whether focus is improving, declining, or remaining stable, providing users with feedback on their progress.

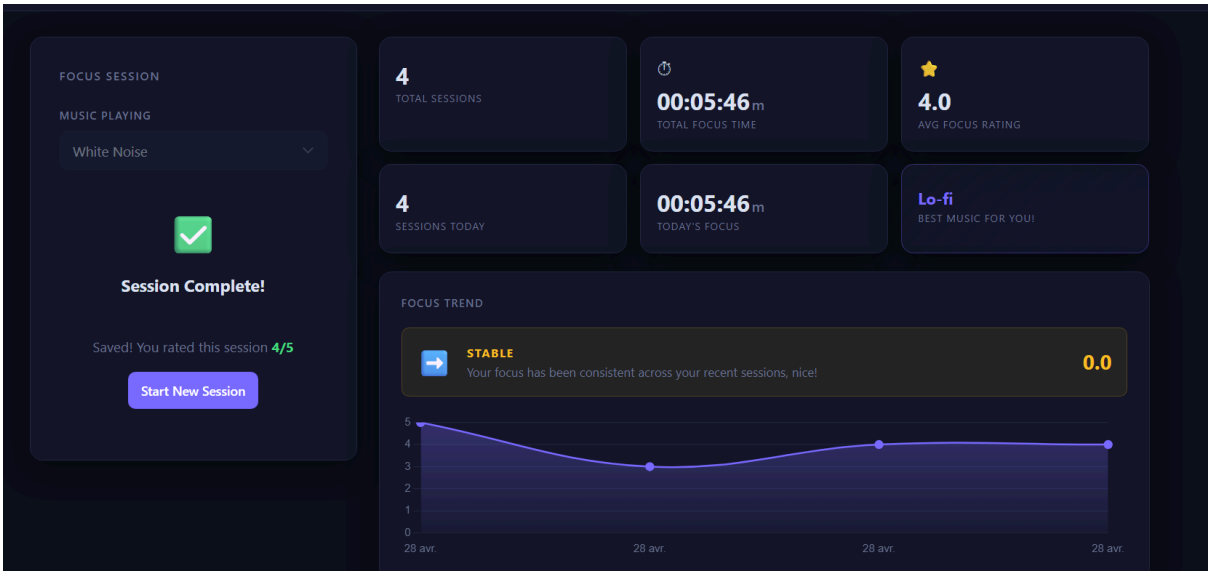


Figure 6: Data Visualization (Chart)
The application uses charts to represent average focus levels across different music types. This visual representation allows users to easily compare performance and identify patterns

in their productivity.



Coda provides a clear, interactive interface to help users track, analyze, and improve their focus during study sessions.

Conclusion:

This project successfully achieved its aim of developing a music-based productivity tracker that analyzes how different types of music influence user focus during study sessions. By integrating concepts from software development, data analysis, and user interaction, the system provides a structured and practical solution to a common problem faced by many students and individuals who rely on music while working. The project demonstrates how technology can be used not only to support productivity, but also to better understand the factors that influence it.

The application allows users to conduct timed focus sessions, select a preferred music type, and evaluate their concentration level using a rating system. This information is then processed to generate meaningful insights, including average focus levels, most effective music categories, and performance trends over time. By transforming subjective experiences into measurable data, the system enables users to make more informed decisions about their study habits. This shift from intuition-based decision-making to data-driven evaluation is a key strength of the system.

Throughout the development process, a structured software engineering approach was followed, including system analysis, design, implementation, and testing. Each stage contributed to the creation of a functional and reliable system. The use of a modular, component-based architecture ensured that the application remained organized, maintainable, and scalable. Additionally, the use of local storage allowed for efficient data persistence without the need for external databases, simplifying the system while maintaining functionality. This design choice was particularly appropriate for the scope of the project, as it reduced complexity while still achieving the intended objectives.

The implementation of data analysis features, such as average calculations and trend evaluation, adds significant value to the application. Rather than simply tracking sessions, the system actively interprets user data to provide personalized feedback. This analytical capability distinguishes the system from basic productivity tools, as it offers users actionable insights rather than raw data. The inclusion of visual elements, such as charts and trend indicators, further enhances usability by presenting complex information in a clear and accessible manner.

From an evaluation perspective, the system performed effectively during testing, successfully capturing session data and generating consistent results. The findings supported existing research, particularly in demonstrating that non-lyrical and low-distraction audio tends to support better concentration. This alignment between the system's outputs and established literature reinforces the validity of the approach taken in this project.

Despite its strengths, the project does have certain limitations. The system relies on self-reported focus ratings, which introduces a degree of subjectivity into the data. Additionally, the use of local storage limits the application to a single user per device and does not support data synchronization across multiple devices. The sample size used during testing was relatively small, which may affect the generalizability of the findings. Furthermore, external factors such as environment, task difficulty, and user mood were not controlled, which may have influenced the results.

There are several potential improvements that could be implemented in future development. Integrating a database system would allow for multi-user support and secure data storage across devices, significantly expanding the system's capabilities. Additional features, such as automated music playback integration (e.g., linking with streaming platforms), more advanced analytics, and personalized recommendations, could further enhance the user experience. Expanding the system to include additional variables, such as task type, time of day, or environmental conditions, could also provide deeper insights into productivity patterns.

Furthermore, future versions of the system could incorporate machine learning techniques to provide more intelligent and adaptive recommendations based on user

behavior over time. This would allow the system to move beyond basic analysis and towards predictive functionality, offering users even greater value.

Finally, the project demonstrates the effectiveness of combining software development with data-driven analysis to address real-world challenges. The system not only provides a functional productivity tool but also highlights the potential of using simple data collection methods to generate valuable insights. Overall, the project meets its pre-defined objectives and serves as a strong foundation for future development and exploration in this area. It also reflects the importance of applying theoretical research to practical applications, bridging the gap between academic knowledge and real-world use.

Video Demonstration:

You may find the link to the video demo here:

https://youtu.be/nbjmspLJS_E

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